

Exercise #2

23. January 2023

Problem 1. (Polynomial interpolation)

You are allowed (but not required) to use python for this problem. Approximate answers obtained by numerical methods are accepted. Consider the function

$$f(x) = 3^{-(x-1)^2}.$$

- a) Interpolate $f(x)$ by a polynomial of minimal degree, by using the interpolation points

$$x_0 = -1, \quad x_1 = 0, \quad x_2 = 1, \quad x_3 = 2$$

What is the maximal error on the intervals $[-1, 2]$ and $[-5, 5]$? Plot $f(x)$ and the interpolating polynomial in the same plot, on the interval $[-1, 2]$.

- b) The *Gaussian nodes* on the interval $[-1, 1]$ are defined, for $i = 0, \dots, n$, as

$$\tilde{x}_0 = \sqrt{\frac{3}{5} - \frac{2}{7}\sqrt{\frac{6}{5}}}, \quad \tilde{x}_1 = -\sqrt{\frac{3}{5} - \frac{2}{7}\sqrt{\frac{6}{5}}}, \quad \tilde{x}_2 = \sqrt{\frac{3}{5} + \frac{2}{7}\sqrt{\frac{6}{5}}}, \quad \tilde{x}_3 = -\sqrt{\frac{3}{5} + \frac{2}{7}\sqrt{\frac{6}{5}}}$$

and are transported on a general interval $[a, b]$ with the transformation rule:

$$x_i = \frac{1}{2}(a+b) + \frac{1}{2}(b-a)\tilde{x}_i.$$

Find the Gaussian nodes on the interval $[-1, 2]$ for $n = 3$.

- c) Interpolate $f(x)$ in the Gaussian nodes for $n = 3$. What is the maximal error on the intervals $[-1, 2]$ and $[-5, 5]$? Plot $f(x)$ and the interpolating polynomial in the same plot, on the interval $[-1, 2]$.
- d) Plot the error as a function in x , for both the interpolation in a) and the interpolation in c), on the interval $[-1, 2]$. Plot both errors in the same plot.

Problem 2. (Lagrange interpolation)

In the lectures and in Maths 3, you have seen how interpolation polynomials can be constructed by solving a system of linear equations. In this exercise, we will discuss an alternative, direct approach, called *Lagrange interpolation*.

(See the note `Introduction.pdf`, Section 2.1 for more information about that approach.)

Given $n + 1$ distinct points x_i , $i = 0, \dots, n$, we define the associated *cardinal functions* as

$$\ell_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} = \frac{x - x_0}{x_i - x_0} \cdots \frac{x - x_{i-1}}{x_i - x_{i-1}} \cdot \frac{x - x_{i+1}}{x_i - x_{i+1}} \cdots \frac{x - x_n}{x_i - x_n}, \quad i = 0, \dots, n.$$

- a) Compute the cardinal functions $\ell_0(x)$, $\ell_1(x)$, $\ell_2(x)$ explicitly for the points $x_0 = -1$, $x_1 = 1$, $x_2 = 5$.
- b) Verify that the cardinal functions satisfy the equations

$$\begin{aligned} \ell_i(x_i) &= 1 && \text{for all } i = 0, \dots, n, \\ \ell_i(x_j) &= 0 && \text{for all } i, j = 0, \dots, n \text{ with } i \neq j. \end{aligned}$$

- c) Assume we are given points $(x_i, y_i)_{i=0}^n$, with distinct values x_i . Show that the interpolation polynomial p_n is given by

$$p_n(x) = \sum_{i=0}^n y_i \ell_i(x).$$

(Hint: Verify that each ℓ_i is a polynomial of degree n and that the interpolation condition $p_n(x_i) = y_i$ is satisfied.)

Problem 3. (Trapezoidal rule (+ adaptivity))

We consider in this problem the numerical integration of the function $f(x) = \sqrt{x}$ on the interval $I = [0, 2]$.

- a) Implement the composite trapezoidal rule for the numerical computation of this integral.
- b) Estimate the numerical converge rate for this method. That is, assume that the error satisfies

$$\text{err}_m := \left| T_m(0, 1) - \int_0^1 \sqrt{x} dx \right| \approx Ch^p,$$

and estimate numerically the number p . (Your estimated p is not necessarily an integer!)

- c) Explain why your result from part b) does not contradict the usual theory of the trapezoidal rule.
- d) Suppose an adaptive trapezoidal rule is implemented. Then, for every sub-interval $[x_i, x_{i+1}]$ two estimates are compared: the estimate for the single interval and the one for a (possible) splitted interval. They are, respectively:

$$T_1 = \frac{x_{i+1} - x_i}{2} (f(x_i) + f(x_{i+1})), \quad T_2 = \frac{x_{i+1} - x_i}{4} \left(f(x_i) + 2f\left(\frac{x_i + x_{i+1}}{2}\right) + f(x_{i+1}) \right)$$

The estimated error difference is computed as $\epsilon = |T_1 - T_2|/3$. If $\epsilon > \text{tol}$, where tol is a desired tolerance, the interval is splitted and the procedure repeated in every sub-interval until the criteria $\epsilon < \text{tol}$ is satisfied. Suppose an initial uniform grid with $m = 10$ (9 intervals) and check (manually or numerically) which intervals would need at least a first refinement with $\text{tol} = 10^{-4}$.

Problem 4. (Runge phenomenon)

You are allowed (but not required) to use python for this problem. Approximate answers obtained by numerical methods are accepted. Consider the function

$$f(x) = \frac{1}{1 + 25x^2}.$$

- a) Interpolate $f(x)$ by a polynomial of minimal degree, by using the interpolation points

$$x_0 = -1, \quad x_1 = 0, \quad x_2 = 1$$

What is the maximal error on the intervals $[-1, 1]$? Plot $f(x)$ and the interpolating polynomial in the same plot, on the interval $[-1, 1]$.

b) The *Chebyshev nodes* on an interval $[a, b]$ are defined as

$$\tilde{x}_i = \frac{a+b}{2} + \frac{b-a}{2} \cos\left(\frac{(2i+1)\pi}{2(n+1)}\right), \quad i = 0, \dots, n.$$

Interpolate $f(x)$ in the Chebyshev nodes for $n = 2$. What is the maximal error on the interval $[-1, 1]$? Plot $f(x)$ and the interpolating polynomial in the same plot, on the interval $[-1, 1]$.

- c) Repeat the analysis of a) and b) (numerically) for $n = 15$ and plot the results.
- d) Plot also the two errors in the same graph and observe the Runge phenomenon for high order degree polynomials and how the Chebyshev nodes can solve the problem.

Problem 5.

We consider the cubic splines $\mathcal{S}_{3,\Delta}$ with knot vector $\Delta = [-2, -1, 0, 1, 2]$.

Which of the following functions is in the space $\mathcal{S}_{3,\Delta}$?

State a reason for each of the functions.

a) $f_1(x) = 2x,$

b) $f_2(x) = |x - 1|^3 + |x + 1|^3,$

c) $f_3(x) = 3(x)_+^2 + (x+1)_+^3,$ where $(y)_+ = \begin{cases} y & \text{if } y \geq 0 \\ 0 & \text{else.} \end{cases}$

d) For some $a, b \in \mathbb{R}$ we consider the function $f_{a,b}(x) = \begin{cases} x^3 + x & \text{for } x \in [-2, 0], \\ x^3 - ax + b & \text{for } x \in [0, 2]. \end{cases}$

Determine for which $a, b \in \mathbb{R}$ this function is a spline in the vector space $\mathcal{S}_{3,\Delta}$ and for which it is not.

The next exercises are optional and should not be handed in! Problem 6. (Inverse interpolation)

We are given a function $f(x)$ with a root r in an interval $[a, b]$, that is, $f(r) = 0$. Moreover,

we assume that the interval is chosen sufficiently small such that the function f is either increasing on the whole interval, or decreasing on the whole interval. We want to find r by using interpolation. To that end, we compute an interpolating function $p(y)$ through the points $(f(x_i), x_i)_{i=0}^n$, for nodes $x_i \in [a, b]$. Then $p(0)$ should be an approximation of r .

- a) Let $f(x) = x^3 - 7$ and $[a, b] = [1.5, 2.0]$. Choose the nodes $x_i = (1.5, 1.75, 2.0)$ and compute an interpolation polynomial in order to approximate the root r . How close are you to the actual root?
- b) Repeat the example with $n + 1$ equidistant nodes over the interval $[a, b]$. Choose $n = 2, 4, 8$, and 16 , find the approximation in each case as well as the error.
- c) Repeat the example with spline interpolation.

Problem 7. (Midpoint rule)

We consider in this problem the numerical integration of the function $f(x) = \log(x)$ on the interval $I = [0, 1]$.

- a) Implement the composite midpoint rule for the numerical computation of this integral.
- b) Estimate the numerical converge rate for this method. That is, assume that the error satisfies

$$\text{err}_m := \left| M_m(0, 1) - \int_0^1 \log(x) dx \right| \approx Ch^p,$$

and estimate numerically the number p .

Explain why your result from part [b](#)) does not contradict the usual theory of the midpoint rule.